

Module 6: Star Schema & Snowflake Schema

Overview

A well-designed data model is the backbone of every successful Power BI solution. While relationships connect tables, dimensional modeling determines how those tables should be organized for maximum performance, scalability, and ease of analysis.

In this module, you will learn the two most common data warehouse modeling techniques:

- Star Schema
- Snowflake Schema

You will also learn Fact Tables, Dimension Tables, Surrogate Keys, Conformed Dimensions, and industry best practices used in enterprise Business Intelligence solutions.

Learning Objectives

After completing this module, you will be able to:

- Understand Dimensional Modeling
 - Design Star Schema models
 - Design Snowflake Schema models
 - Differentiate Fact and Dimension Tables
 - Understand Surrogate Keys
 - Understand Conformed Dimensions
 - Build enterprise-grade Power BI models
 - Improve report performance through proper modeling
-

Table of Contents

1. Introduction to Dimensional Modeling
2. Why Schema Design Matters
3. Fact Tables
4. Dimension Tables
5. Measures vs Dimensions
6. Star Schema
7. Components of Star Schema
8. Benefits of Star Schema
9. Snowflake Schema
10. Components of Snowflake Schema
11. Benefits & Drawbacks of Snowflake Schema
12. Star Schema vs Snowflake Schema

13. Surrogate Keys
14. Natural Keys
15. Conformed Dimensions
16. Slowly Changing Dimensions (SCD)
17. Date Dimension
18. Best Practices
19. Common Modeling Mistakes
20. Key Terminologies
21. Practice Exercises
22. Mini Project
23. Module Summary

Introduction to Dimensional Modeling

What is Dimensional Modeling?

Dimensional Modeling is a data modeling technique used in Business Intelligence and Data Warehousing.

Its goal is to organize data for:

- Faster querying
- Easier reporting
- Better performance
- Simpler analysis

Traditional Database Design

Transactional databases are designed for:

- Insert Operations
- Update Operations
- Delete Operations

Examples:

- Banking Systems
- ERP Systems
- CRM Systems

Analytical Database Design

Analytical systems are designed for:

- Reporting
- Dashboards
- Business Analysis

Power BI works best with analytical models.

2 Why Schema Design Matters

Poor schema design causes:

- Slow reports
 - Complex DAX
 - Incorrect aggregations
 - Difficult maintenance
-

Good Schema Design Provides

Better Performance

Queries execute faster.

Simpler Relationships

Fewer joins required.

Easier Analysis

Users understand the model quickly.

Better Scalability

Supports growing datasets.

3 Fact Tables

What is a Fact Table?

A Fact Table stores measurable business events.

Facts are typically numeric values used for analysis.

Characteristics

- Contains measurements
- Usually very large
- Connected to multiple dimensions
- Located at the center of the model

Example

FactSales

OrderID	ProductID	CustomerID	DateID	SalesAmount
1	P101	C101	D001	5000

Common Fact Table Metrics

- Revenue
 - Profit
 - Quantity Sold
 - Cost
 - Discount
 - Transactions
-



Dimension Tables

What is a Dimension Table?

Dimension Tables contain descriptive attributes about business entities.

Characteristics

- Descriptive information
 - Smaller than fact tables
 - Used for filtering and grouping
-

Example

DimProduct

ProductID	ProductName	Category
P101	Laptop	Electronics

Common Dimension Tables

Customer Dimension

Contains:

- Customer Name
- Gender

- Age
 - City
-

Product Dimension

Contains:

- Product Name
 - Brand
 - Category
-

Date Dimension

Contains:

- Year
 - Quarter
 - Month
 - Week
-

5 Measures vs Dimensions

Understanding the difference is critical.

Measures

Numeric values that can be aggregated.

Examples:

```
Revenue
Profit
Sales
Quantity
```

Dimensions

Descriptive fields used to categorize measures.

Examples:

```
Product
Region
```

Customer
Date

Example

Product	Revenue
Laptop	5000

Product = Dimension

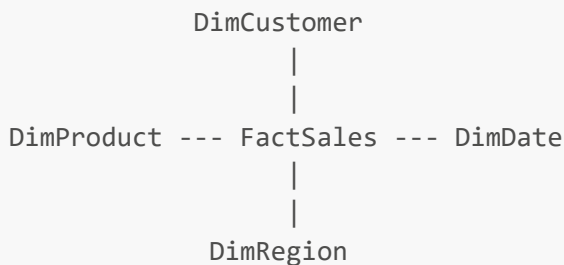
Revenue = Measure

6 Star Schema

What is a Star Schema?

A Star Schema is a dimensional model where a central fact table connects directly to multiple dimension tables.

Structure



Characteristics

- Fact table in center
 - Dimensions around fact table
 - Simple relationships
 - Most recommended design
-

7 Components of Star Schema

Fact Table

Stores business transactions.

Example:

FactSales

ProductID	CustomerID	Sales
P101	C101	5000

Dimension Tables

DimCustomer

CustomerID	Name
C101	Rahul

DimProduct

ProductID	ProductName
P101	Laptop

DimDate

DateID	Month
D001	January



Benefits of Star Schema

Better Performance

Requires fewer joins.

Easier Maintenance

Simple structure.

Better DAX Performance

Optimized filter propagation.

Easier Reporting

Business users understand the model.

Why Power BI Prefers Star Schema

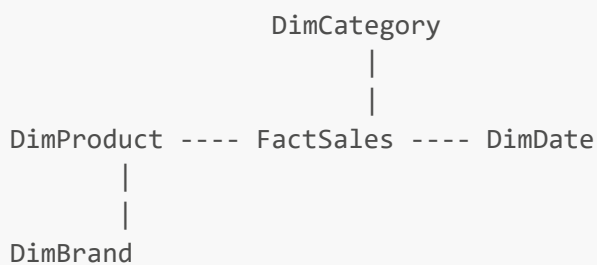
Power BI's VertiPaq Engine performs best with star schemas.

9 Snowflake Schema

What is a Snowflake Schema?

A Snowflake Schema is an extension of Star Schema where dimension tables are normalized into additional tables.

Structure



Characteristics

- Normalized dimensions
 - More relationships
 - Reduced data redundancy
-

10 Components of Snowflake Schema

Fact Table

Same as Star Schema.

Dimension Tables

Split into smaller related tables.

Example

Product Table

ProductID	BrandID	CategoryID
P101	B1	C1

Brand Table

BrandID	BrandName
B1	Dell

Category Table

CategoryID	Category
C1	Electronics

1

1

 Benefits & Drawbacks of Snowflake Schema

Benefits

Reduced Redundancy

Data stored once.

Better Data Integrity

Consistent information.

Smaller Dimension Tables

Less duplication.

Drawbacks

More Complex

Harder to understand.

More Relationships

Additional joins required.

Slower Queries

Compared to Star Schema.

1 2 Star Schema vs Snowflake Schema

Feature	Star Schema	Snowflake Schema
Structure	Denormalized	Normalized
Complexity	Simple	Complex
Performance	Faster	Slower
Relationships	Fewer	More
Maintenance	Easier	Harder
Power BI Recommendation	Preferred	Less Preferred

1 3 Surrogate Keys

What is a Surrogate Key?

A system-generated unique identifier.

Example

CustomerKey	CustomerID
1	C101
2	C102

Benefits

- Stable identifiers
 - Improved performance
 - Simplified relationships
-

1 4 Natural Keys

What is a Natural Key?

A key that already exists in business data.

Example

CustomerID

C101

CustomerID is a natural key.

Problems

Business identifiers may change over time.

1 5 Conformed Dimensions

What is a Conformed Dimension?

A dimension shared across multiple fact tables.

Example

Date Dimension

Used by:

- Sales Fact
 - Inventory Fact
 - Finance Fact
-

Benefits

Provides consistency across reports.

1 6 Slowly Changing Dimensions (SCD)

What is SCD?

Dimension attributes change over time.

Example

Customer moves:

Delhi → Mumbai

SCD Type 1

Overwrite old value.

SCD Type 2

Keep historical records.

SCD Type 3

Store limited history.

17

Date Dimension

A Date Table is one of the most important dimensions.

Example

Date	Year	Quarter	Month
01-Jan-2026	2026	Q1	January

Why Important?

Required for:

- Time Intelligence
- YTD Analysis
- MTD Analysis
- YoY Comparison

Typical Columns

- Date
- Day
- Week
- Month
- Quarter
- Year

- Fiscal Year

1 8 Best Practices

Use Star Schema

Preferred for Power BI.

Keep Dimensions Descriptive

Store attributes in dimensions.

Keep Facts Numeric

Store measurements in facts.

Create Date Dimension

For all reporting models.

Avoid Many-to-Many Relationships

Whenever possible.

Use Meaningful Naming

Examples:

```
FactSales
DimCustomer
DimProduct
DimDate
```

1 9 Common Modeling Mistakes

Using One Massive Table

Creates poor performance.

Ignoring Date Dimension

Limits Time Intelligence.

Excessive Snowflaking

Adds complexity.

Duplicate Dimensions

Creates inconsistency.

Incorrect Relationships

Causes reporting errors.

 Key Terminologies

Term	Definition
Fact Table	Stores measurable data
Dimension Table	Stores descriptive attributes
Star Schema	Central fact with surrounding dimensions
Snowflake Schema	Normalized dimensions
Surrogate Key	System-generated identifier
Natural Key	Business-generated identifier
Conformed Dimension	Shared dimension across facts
SCD	Slowly Changing Dimension
Denormalization	Combining tables
Normalization	Splitting tables

 Practice Exercises

Exercise 1

Identify Fact and Dimension tables from a retail dataset.

Exercise 2

Design a Star Schema for:

- Customers
 - Products
 - Sales
 - Dates
-

Exercise 3

Convert a Star Schema into a Snowflake Schema.

Exercise 4

Create a Date Dimension table.

Exercise 5

Compare Star and Snowflake Schema advantages.



Mini Project

Retail Data Warehouse Model

Tables

FactSales

SaleID	CustomerID	ProductID	DateID	Revenue
--------	------------	-----------	--------	---------

DimCustomer

CustomerID	CustomerName
------------	--------------

DimProduct

ProductID	ProductName
-----------	-------------

DimDate

DateID	Month	Year
--------	-------	------

Tasks

1. Build Star Schema

2. Create Relationships
3. Verify Cardinality
4. Create Sales Report
5. Analyze Performance



Module Summary

In this module, you learned:

- ✓ Dimensional Modeling
- ✓ Fact Tables
- ✓ Dimension Tables
- ✓ Measures vs Dimensions
- ✓ Star Schema
- ✓ Snowflake Schema
- ✓ Surrogate Keys
- ✓ Natural Keys
- ✓ Conformed Dimensions
- ✓ Slowly Changing Dimensions
- ✓ Date Dimensions
- ✓ Enterprise Modeling Best Practices

A properly designed Star Schema is one of the most important factors in creating fast, scalable, and maintainable Power BI solutions.



Next Module

Module 7: Data Visualization Fundamentals

Topics Covered:

- Principles of Data Visualization
- Data Storytelling
- Choosing the Right Chart
- Visual Perception
- Dashboard Design Fundamentals
- Color Theory
- User Experience (UX) in Reporting
- Visualization Best Practices